COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	17/07/2026	**DATE SUBMITTED:**	17/07/2026
NAME:	Bungag, Marc Jhowell M.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

HOA 2.1 Array, Pointers and Dynamic Memory Allocation

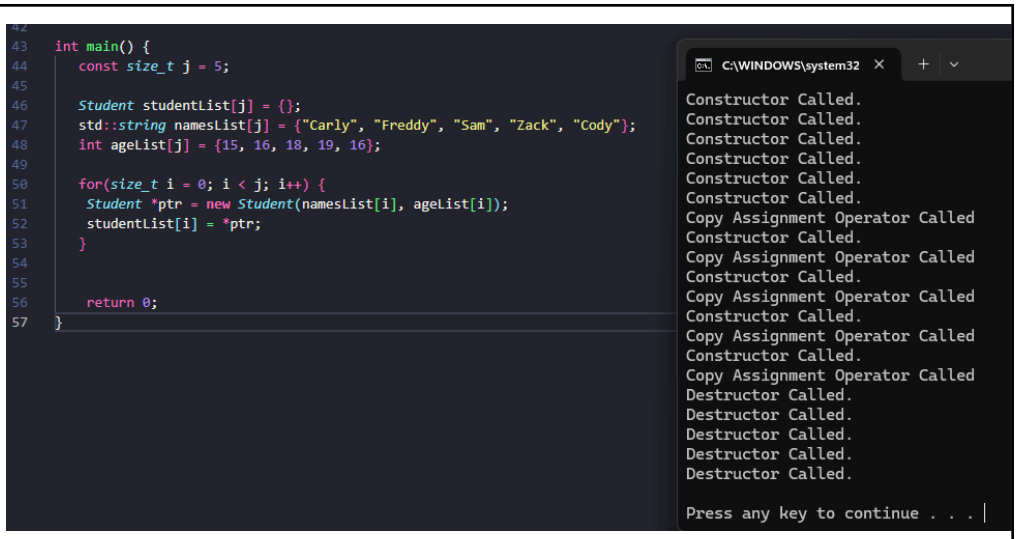
1. PROCEDURE OUTPUT

What procedures did you perform?

Screenshot	
Observation	In this code, I observed that the code provided did not have a copy assignment operator, so I added a copy assignment operator in order for the “student3” to work that is why the output has “Copy Assignment Operator Called”. When I run the code without

	the copy assignment, it still has an output but it has a warning that the student3 is not working or cannot copy the student2.

Screenshot	
Observation	The code is still the same and it does not require any modification in the class. However, there is an array provided with the name and age of the student. It creates 5 objects. It does not print the expected names and corresponding age since it has no setter function for the nameList and ageList. Hence, the output presented that the constructors and destructors are called.

Loop A	
Observation	In this code I added a function in order for the nameList and ageList to work, however, it still printed the calling of constructors, destructors, and copy assignment operators. This is a good indicator that the code is working properly and it just needs to print the details of each person. The modification states that, for each student (i) printed is less than j (which is 5), a student must be printed out again to reach the end of the loop.
Loop B	
Observation	Now we printed the “.printDeatils” in order to show each student's name and their corresponding age. The same loop is applied to match the given number of names and ages in the array.

Output

ILOA.cpp > main()

```cpp
#include <iostream>
#include <string.h>

class Student{
private:
    std::string studentName;
    int studentAge;

public:
    //constructor
    Student(std::string newName = "John Doe", int newAge = 18) {
        studentName=std::move(newName);
        studentAge = newAge;
        std::cout << "Constructor Called."<<"\n";
    };

    //deconstrcutor
    ~Student() {
        std::cout<<"Destructor Called."<<"\n";
    }

    //Copy Constructor
    Student(const Student &copyStudent){
        std::cout<< "Copy Constructor Called."<<"\n";
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
    }

    //Copy Assignment
    Student& operator=(const Student& copyStudent){
        std::cout<<"Copy Assignment Operator Called"<<"\n";
        studentName = copyStudent.studentName;
        studentAge = copyStudent.studentAge;
        return *this;
    }

    //Display Attribute
    void printDetails(){
        std::cout<<this->studentName<< " " <<this->studentAge<<"\n";
    }
};

int main() {
   const size_t j = 5;

   Student studentList[j] = {};
   std::string namesList[j] = {"Carly", "Freddy", "Sam", "Zack", "Cody"};
   int ageList[j] = {15, 16, 18, 19, 16};
```

```
49
50      for(size_t i = 0; i < j; i++) {
51       Student *ptr = new Student(namesList[i], ageList[i]);
52       studentList[i] = *ptr;
53      }
54
55      for(size_t i = 0; i < j; i++){
56       studentList[i].printDetails();
57      }
58
59       return 0;
60  }
```

```
C:\WINDOWS\system32
Copy Assignment Operator Called
Constructor Called.
Copy Assignment Operator Called
Constructor Called.
Copy Assignment Operator Called
Constructor Called.
Copy Assignment Operator Called
Carly 15
Freddy 16
Sam 18
Zack 19
Cody 16
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.

Press any key to continue . . .
```

Observation	In this output, we finally printed out the names and ages of the arrays. Each name has a corresponding age. The loop was successful and the arrays were traversed.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

ILOA.cpp | ILOC.cpp

ILOC.cpp > ...

```cpp
#include <iostream>
#include <string>

class Groceries {
private:
    std::string type;
    std::string name;
    float price;
    int quantity;

public:
    // Constructor
    Groceries(std::string T = "", std::string N = "", double P = 0, int Q = 0) {
        type = T;
        name = N;
        price = P;
        quantity = Q;
    }

    // Destructor
    ~Groceries() {
        std::cout << name << " has been bagged." << std::endl;
    }

    // Copy Constructor
    Groceries(const Groceries &copyGroceries) {
        type = copyGroceries.type;
        name = copyGroceries.name;
        price = copyGroceries.price;
```

Ln 59, Col 1 Spa

ILOC.cpp > ...

```
 4  class Groceries {
26      Groceries(const Groceries &copyGroceries) {
29          price = copyGroceries.price;
30          quantity = copyGroceries.quantity;
31      }
32
33      // Copy Assignment Operator
34      Groceries& operator=(const Groceries& copyGroceries) {
35          if (this != &copyGroceries) {
36              type = copyGroceries.type;
37              name = copyGroceries.name;
38              price = copyGroceries.price;
39              quantity = copyGroceries.quantity;
40          }
41          return *this;
42      }
43
44      // Calculate total price
45      double calculateSum() {
46          return price * quantity;
47      }
48
49      // Display item details
50      void ScannedItems() {
51          std::cout << "Type: " << type << std::endl;
52          std::cout << "Name: " << name << std::endl;
53          std::cout << "Price: PHP " << price << std::endl;
54          std::cout << "Quantity: " << quantity << std::endl;
55          std::cout << "Total: PHP " << calculateSum() << std::endl;
```

Ln 59, C

ILOC.cpp > ...

```
 4  class Groceries {
50      void ScannedItems() {
56          std::cout << std::endl;
57      }
58  };
59
60  float TotalSum(Groceries* GroceryList[], int groceries) {
61      float total = 0;
62
63      for (int i = 0; i < groceries; i++) {
64          if (GroceryList[i] != nullptr) {
65              total += GroceryList[i]->calculateSum();
66          }
67      }
68      return total;
69  }
70
71  int main() {
72      Groceries* GroceryList[4];
73
74      GroceryList[0] = new Groceries("Fruit", "Apple", 10, 7);
75      GroceryList[1] = new Groceries("Fruit", "Banana", 10, 8);
76      GroceryList[2] = new Groceries("Vegetable", "Broccoli", 60, 12);
77      GroceryList[3] = new Groceries("Vegetable", "Lettuce", 50, 10);
78
79      std::cout << "~~~~~Jenna's Grocery List~~~~~"<<std::endl;
80
81      for (int i = 0; i < 4; i++) {
82          GroceryList[i]->ScannedItems();
```

Ln 59, Co

ILOA.cpp | ILOC.cpp

ILOC.cpp > ...

```cpp
float TotalSum(Groceries* GroceryList[], int groceries) {

int main() {
    Groceries* GroceryList[4];

    GroceryList[0] = new Groceries("Fruit", "Apple", 10, 7);
    GroceryList[1] = new Groceries("Fruit", "Banana", 10, 8);
    GroceryList[2] = new Groceries("Vegetable", "Broccoli", 60, 12);
    GroceryList[3] = new Groceries("Vegetable", "Lettuce", 50, 10);

    std::cout << "~~~~~Jenna's Grocery List~~~~~"<<std::endl;

    for (int i = 0; i < 4; i++) {
        GroceryList[i]->ScannedItems();
    }

    std::cout << "Total Cost: PHP " << TotalSum(GroceryList, 4) << std::endl;

    // Delete Lettuce
    delete GroceryList[3];
    GroceryList[3] = nullptr;
    std::cout << "After deleting Lettuce:" << std:: endl;
    std::cout << "Updated Total Cost: PHP " << TotalSum(GroceryList, 4) << std::endl;

    delete GroceryList[0];
    delete GroceryList[1];
    delete GroceryList[2];

    return 0;
}
```

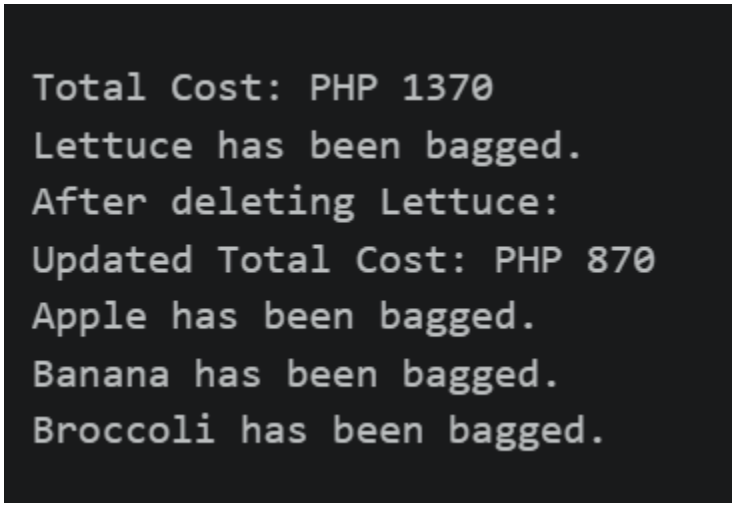
Ln 59, Col 1

```
~~~~~Jenna's Grocery List~~~~~
Type: Fruit
Name: Apple
Price: PHP 10
Quantity: 7
Total: PHP 70

Type: Fruit
Name: Banana
Price: PHP 10
Quantity: 8
Total: PHP 80

Type: Vegetable
Name: Broccoli
Price: PHP 60
Quantity: 12
Total: PHP 720

Type: Vegetable
Name: Lettuce
Price: PHP 50
Quantity: 10
Total: PHP 500
```



In this code I named the class “Groceries” and declared all the parameters needed like the type, name, price, and quantity of the vegetable in private. In the public, I put a constructor, destructor, copy constructor, and copy assignment operator to declare each groceries and will be printed out in the main driver. Then, I created an equation to solve for the sum of groceries scanned, and made a void function to print out the parameters. Lastly in the main driver, all groceries are then printed out with their corresponding type, price, and quantity. The lettuce is then deleted and the total cost of the groceries are then printed out. After printing the groceries, they are then deleted.

In this lesson, I had a review about the class in c++. I learned little about it when I was in my 1st year 1st semester. This data type is used to collect the parameters and function into a single usable package. This is a way to organize the code and to minimize function calling and lines in the code.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.’s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance

with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.